

Heaven's Light is Our Guide



Rajshahi University of Engineering and Technology

Department of Electrical and Computer Engineering

Report 01

Course Title : Digital Techniques - Sessional

Course No: ECE 2112

Submission Date: 25.07.26

Submitted by
Shakib Shadman
2410025

Submitted to
Mst. Mazeda Noor Tasnim
Lecturer
Electrical & Computer Engineering,
RUET

Experiment No-01

Problem Definition

Design and implement an **AND gate using only NAND gates**. Verify the circuit by comparing the output with the corresponding truth table.

Circuit Diagram

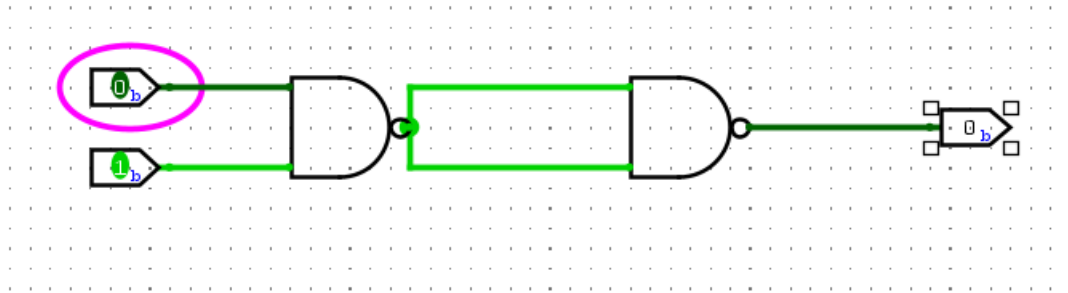


Figure 1: Logic Diagram of AND Gate using NAND Gates

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Expression

$$Y = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$$

Experiment No-02

Problem Definition

Design and implement an **OR gate using only NAND gates**. Verify the output using the truth table.

Circuit Diagram

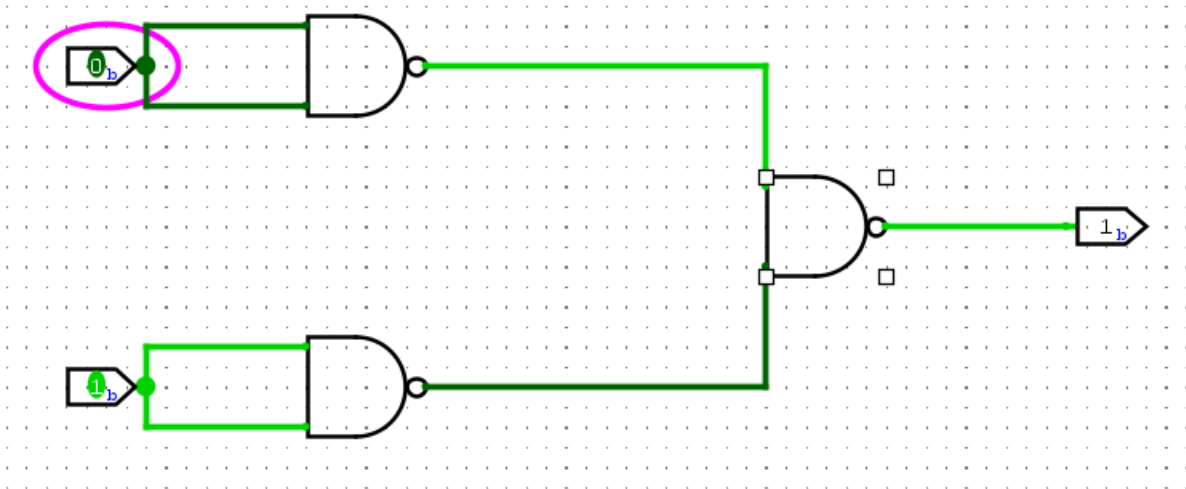


Figure 2: Logic Diagram of OR Gate using NAND Gates

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Expression

$$Y = (A \text{ NAND } A) \text{ NAND } (B \text{ NAND } B)$$

Experiment No-03

Problem Definition

Design and implement a **NOT gate using only NAND gates**. Verify its operation using the truth table.

Circuit Diagram

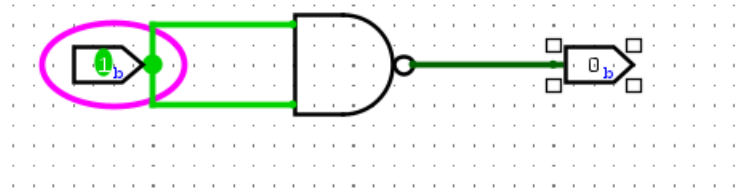


Figure 3: Logic Diagram of NOT Gate using NAND Gate

Truth Table

A	Y
0	1
1	0

Expression

$$Y = A \text{ NAND } A$$

Experiment No-04

Problem Definition

Design and implement an **AND gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

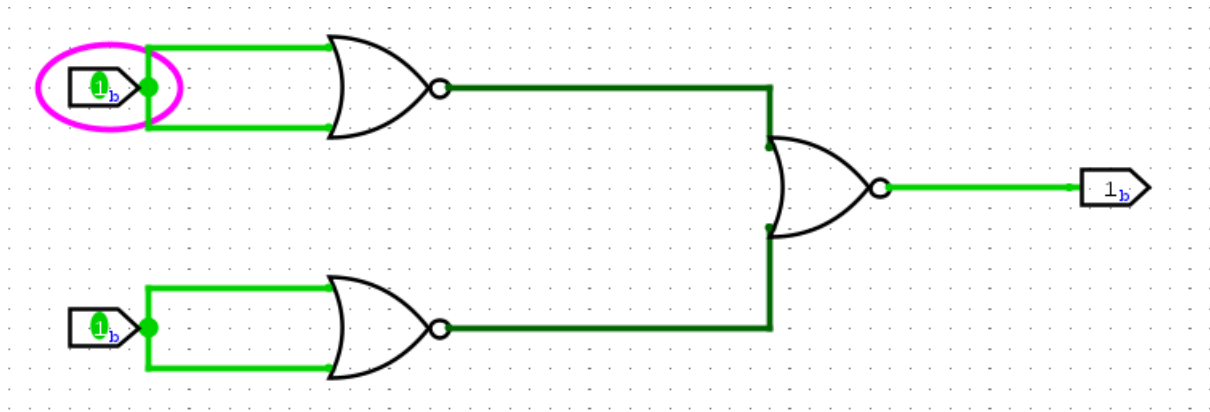


Figure 4: Logic Diagram of AND Gate using NOR Gates

Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Expression

$$Y = (A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$$

Experiment No-05

Problem Definition

Design and implement an **OR gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

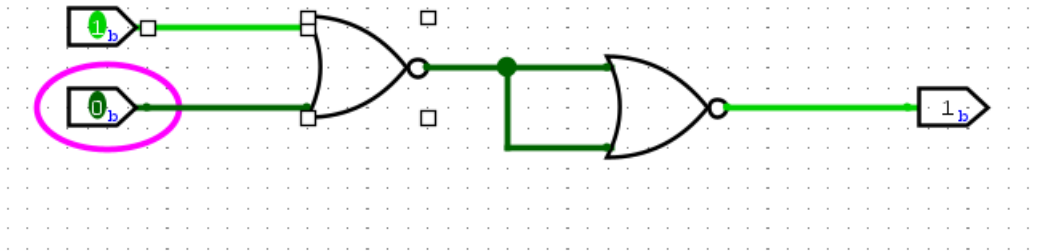


Figure 5: Logic Diagram of OR Gate using NOR Gates

Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Expression

$$Y = (A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$$

Experiment No-06

Problem Definition

Design and implement a **NOT gate using only NOR gates**. Verify the output using the truth table.

Circuit Diagram

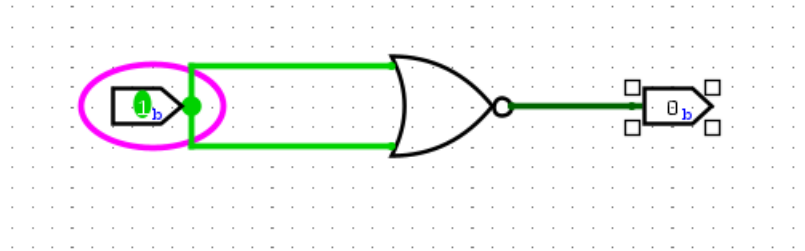


Figure 6: Logic Diagram of NOT Gate using NOR Gate

Truth Table

A	Y
0	1
1	0

Expression

$$Y = A \text{ NOR } A$$

Experiment No-07

Problem Definition

Design and implement a **1-bit Full Adder** using logic gates. Verify the correctness of the circuit by comparing the generated outputs with the theoretical truth table.

Circuit Diagram

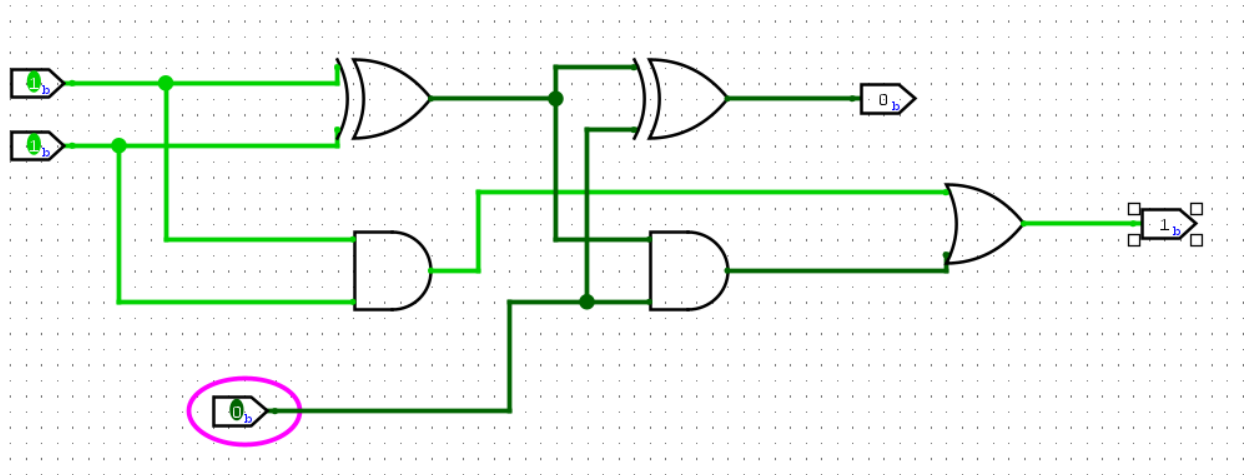


Figure 7: Logic Diagram of 1-Bit Full Adder

Truth Table

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Expressions

$$S = A \oplus B \oplus C_{in}$$
$$C_{out} = AB + C_{in}(A \oplus B)$$

Experiment No-08

Problem Definition

Design and implement a **Binary-to-BCD Converter**. Verify the circuit by applying different binary input combinations and comparing the outputs with the expected BCD values.

Circuit Diagram

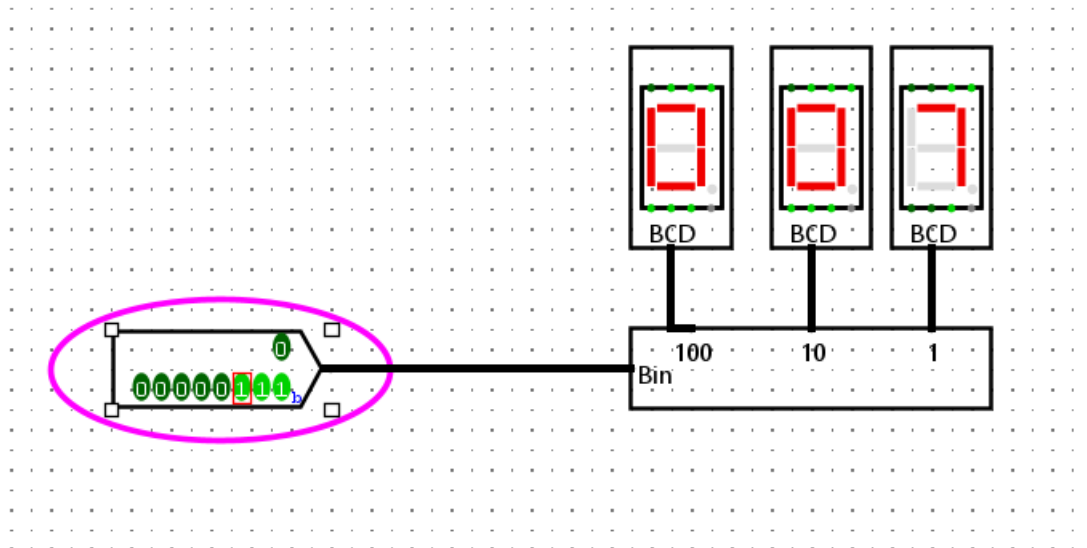


Figure 8: Logic Diagram of Binary-to-BCD Converter

Truth Table

Binary Input	Decimal	Hundreds	Tens	Ones
0000000	0	0	0	0
0000001	1	0	0	1
0000010	2	0	0	2
0000011	3	0	0	3
0000100	4	0	0	4
0000101	5	0	0	5
0000110	6	0	0	6
0000111	7	0	0	7
0001000	8	0	0	8
0001001	9	0	0	9
0001010	10	0	1	0
0001011	11	0	1	1
⋮	⋮	⋮	⋮	⋮
1100100	100	1	0	0
1111111	127	1	2	7

Expression

Binary $\rightarrow$ BCD